

Step4c_DynamicFC and DynamicFC_plot Usage Guide

Step4c_DynamicFC

Overview

Step4c_DynamicFC calculates the dynamic functional connectivity of a participant using ft_connectivityanalysis and sliding window analysis. The function utilizes the beamformed data from Step3. The calculation is performed for every frequency band specified for a particular participant and outputs a .mat file with these values in a matrix that is dimensioned as number of time windows x ROI x ROI x frequency bands and saves some of the parameters for the plotting function.

Usage

The function requires patient ID, visit number and a config structure. The config structure must also include “.meta” path names (or the directory in which the data files are stored in, specifically cfg.meta.project_path and cfg.meta.analysis_name). A sample config structure and function call are shown below:

```
config.step4c.connmethode = 'wpli_debiased';
config.step4c.bandavgmethod = 'max';
config.step4c.toi = [0 1];
config.step4c.winsize = 0.25;
config.step4c.stepsize= 0.25;
config.step4c.freqbands = [1, 3; ...
                           4, 8; ...
                           8, 12];
Step4c_DynamicFC(config, "WMP_19", 1);
```

Limitations/Things to Note

1. The function will cut off data if the window and step size do not perfectly fit in the time of interest. Window and step sizes do not need to be the same.

- a. Examples with no data cut off:

Window and step size are the same:

Window size: 0.25 s

Step size: 0.25 s

Time of interest: 0-1 s

Windows: 0 – 0.25 s, 0.25 – 0.5 s, 0.5 – 0.75 s, 0.75 – 1 s

Window and step size are different:

Window size: 0.2 s

Step size: 0.1 s

Time of interest: 0 – 0.5 s

Windows: 0 – 0.2s, 0.1 – 0.3 s, 0.2 – 0.4s, 0.3 – 0.5 s

- b. Example with data cut off:

Window size: 0.3 s

Step size: 0.3 s

Time of interest: 0 – 1 s

Windows: 0 – 0.3s, 0.3 – 0.6s, 0.6 – 0.9 s (last 0.1 s get cut off)

2. The function averages across trials.

DynamicFC_plot

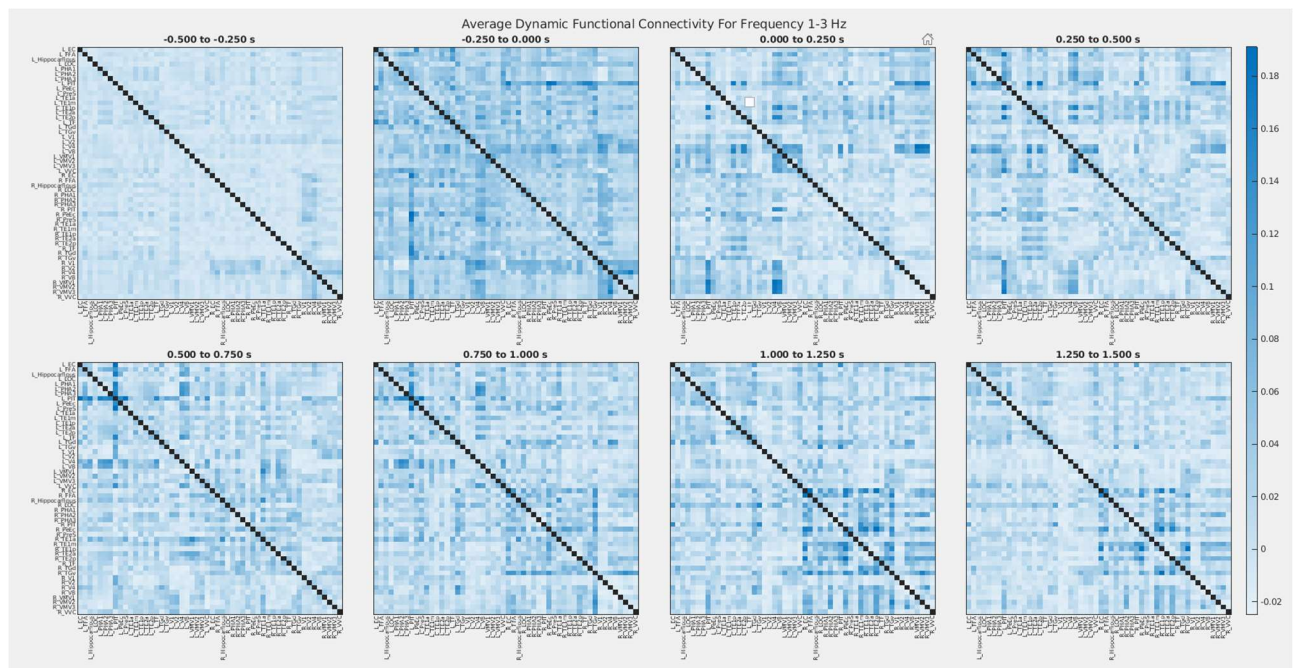
Overview

DynamicFC_plot utilizes the output of Step4c_DynamicFC to plot average ROI x ROI heatmaps over time for different participants and visit numbers. This function can be used with just one participant ID. The function saves the heatmaps for each frequency band and a .mat file containing the average values that were plotted, and the participant IDs and visit numbers used to calculate the average. The heatmaps and .mat files are stored in a group folder if more than one participant ID is given. If only one participant ID is given, it will save in the corresponding session number if only one is given, or in the participant folder if several visit numbers are given.

Usage

The function requires a config structure as an input. A sample config structure and function call is shown below. The config structure must also include “.meta” path names, as described above. In this config structure, the average is taken for session 1 for participant 1, and sessions 1 and 2 for participant 4.

```
config.DFCplot.pids = ["WMP_01" "WMP_04"];  
config.DFCplot.visitnum = {[1] [1 2]};  
config.DFCplot.connmeth = 'wpli_debiased';  
DynamicFC_plot(config);
```



Limitations/Things to Note

1. The function assumes that the data for all participants and visits uses the same connectivity metric, windowing, ROIs, and frequency bands.
2. Plots heatmaps in rows, with at most 4 windows per row.